

21738557678957308158741642336353824280332427554930462924011189994109105105399130047664661471458793302528156074488074768125325552059117162238556309527112834936402188626268571709336
45694202796740283388079705693622513201772597658648181438528963861665998788858535898154438503463306784965798673820160475797806229237168668421717746660061572961764510105103169097177
93218918699931349273467342317922333476864840697820028341727685980423969586730570243991352609264056022150718530023424268517344831235455568266741227960958575481850812348038079463233
63606417211054601458552928534493033118150115619161555693770241770550554499529737582114074182431380120714772840707852159395088269493664238748745057279742776132574469624384875689653
30407412894453489225054090776932666850029595292186134437947662665747297246043430872616042124436533539769799139986823422779653752382963767747267379185734595769915559427673674855429
75370942632242760502881500343343040316034381814730747676072734033077735843254171377377014779718463974046103745567964956757027131034407403708211353374978946864024178311632866586041
72754902404505248797999513552439537747437062838507349149574411692954309118305515784336212767794896245651850429003622360203311195922091308745264247426275664262903368105963447993872
98240422112911361143059016409617867786232863676502373082751858489441126811219236377564464833673632632647963075404039664625212748372821367043432462068005969819856553185661885457484
99974909862369230495951904734936950051872981195356642135768041212088418241018048889389996423725595399850318107769125985769041199264454181737875017650320365142552155760121226514429
24710742174620825852533362603916493693733437565515864571515256852947599965158993347461724161145290991367516751601939399361015051015476243808784998657546465004465440461390458547052
49711460647694045626090672955009187249313882623739879069895915767905896348917566698474916407889816143469512610314907172315706002974660958802140140184715510886560816163901931865860
47699543111936903648647442159874643473523431419360238723051816988561349040182830449009950411547056646457879499863484626974676268081222224957508877304579694468978994440066790339130
72706748099943675543189461141357225260171682472427925018950980984220501517226327607778798851466147373957921044143564455174320126079978097464623700636831069918558462276346545528188
42317165599122529933805589797627367734607655934429275363173054069079037257591512966716245102140337082902843517528205278332938790874214880643420410458491632470415932769257662862696
61433200036549822886252674758485698469708070817533896150105859115642685876654753497556964985788065398637793521816168103262975022073434907535426357686042118451470569394939612967231
30208253304056628652592001346954336454040797972493069703646326671026143247996691554289142216805305419070058131841159213030912196664626769201105184502933216712498716891145667825079
78500777799656219685534970150461019743125646461263065892034787738048072793773583984330409695609507620696803558985641025193721433773240279558227788352119714813805113612801550772312
86849796452340734452637760021665610614974156633474428918742068853077854879652529641758487822636663499143841003487803164447240009919245839323144696580924791519372965682862633247652
9841474514647394071364816823199722920873323701887168167533225983168201088300122761264779075664800830563657951847335951228125174279824737146539718373566040936565184725881675021710
15323691833407859445273899695760317616610066550054893507704309740165339361112925098500181635984168949714344860797354467275286420034379280966907424137737233656089431072730073046780
98880529212371450572710173892711771416123398512327146880992981719140244469582305197245877790941704760120596662120749438717379823540291496732157119647374498142722722739832479240420
83863279274154380509418733001015984358985741224927228265085884438288667739962013337901809725099065245906867431204375489020349595878759526270043987382718764837721117503622974080735
80533085250494756555884164207901134553024717390892060560986023424601314810258154147576776160182453759375471038629505854411483401401132415260872217269603562167442331347758730182463
44414843575208057626937170372968839556041503883654348768734261535699946260426236008851406273533056447159900116073871653500636051042440236901055588086775405834458842110884217320739
20268810173937898670637727684337721813682806507142230622788714185953629286276229330242619504014044379624503675971628399724575929820599897342285046886515922242765628549182509843863
34229811379708990046602140708436729673989461356085606414641929132190493715373721357015069905576209566405283368198033540252540602947064632982146998601832040698208781576361302121562
50755006249707373872896077499889313183557481848243548366873400211908856063916902488620330937416642369505590925184494040949442008804665361903645057710305714731065802391281698094115
43823085909945916334768114352907357042400263855642408474706572169208521291204510159028800098284419824550103703866758193930239827906219738600335670534586258940392205848771393952691
39622463712420432603024720625265383590019797842457896460153222360066127160485665079993833494025974641865409019457042365327683657956383305267707584461111137572775949990253189459080
11614749159025840308915677622503166625783937426109024501670956297265524646401614007091019074891203172657106041258196673126756670837806236322428389866470981436581520921524194069224
50462489892540002936717052922480888327808298523307178077994081014625678528415086267778466947777142080569691094775976156340169402382643718254828470004921141687541022747113123201664
99407783418253513792656275120567447088755266935746250142447174227788655627540812428580233830988280923278424507406042441476844381363437118898923454514775836462919713864399529760717
60747705991622399919951166840510892224922206217778769018870211802215896459890560045240467980162613363631261914081977562598352389903391255120103058406674661603509382374562456358329
20763483566597743214439042428972830283709088454118195354985719097596478635049280448787702028527928700021483384147291766642143539964858651313977216068128261765929808603165226198865
64762530570697034587633809351751832059082487177370771913989929797230878048051868577796837362314420795866224566668859020281722200872556972172358373625984652770785271563678599061762
01060746369639922210287864166264172780580790396131892007224729217608909877808416485010995316693842647864165694700992996957560654703285629253146411664844080679266913381902263984626
64641352558772454524700081493289101439355346755205057827175591852975023524063991704638404750132701489555843939090917905864088513227573234243165486961403870932041006024400710074544
27713923782807914435754437066096150222681015273184062201845581880474232465249303894301602074072010382984784792744784212550560182942332943507268310008778608770415886153638359078685
84557950308234323782891078030901886948676429853206997210740958920428000846144157873728173868950596613028481732581053137797549120538399133616432681221607177385535808930270709308771
54597957548513429979224943670471591611736431439156893966586257189591726800302369525873486217877552057462402722686672320548829570760371863624579563058476705334625815923487374263606
08496330083054243357530484076024679548905450646301199350499418323093900856148479278611688045336000084923102373143142229316431018227984854862961740186145011287616151764079329743688
66346048915073773178090510751182014505528198587572996183048641990174612887697962728937294741654326162116355508791583788581874259852658623769438819004911776943333819190195607105943
32306227007902422911796281609259338544740087186689590731413988938710866996486388699172434968604506839339593308383024608753629397430197493638672845557809973853293500374490230085282
34779895106730200213730174445347376792265388418727705088769838572646154000677030976090511450056287703562697361463847956598188701301311451560340386409417030387113791665950618442953
46848030591626835292798672026537348316677372738859638716750857018141366584097236173859588853933756306149668961778723762384127770207269835973795425048184147079891961362478724229573
53935256228721857329640946630438182620085804642994239128649538594354919348076270262538944319461627741356378515337820986103194078492045231199497606711150257102219353292836955038703
14606709938780192711093909902329555246366201385779607401789770411755600276640049661748425556766125644288332171244214984102887616902558185419611940230527293122853390126909144483704
62502020101281222016224448570256155410995889797939031853092904877616355097298010086162680278536529888026663620463400702084282567393020700047015857299136920402831615377658880291959
86447038982229354030350901129545928453827858356150478866056301283839622254193264759969222756151108290370774433326893982888818447309872524175829885836928973102836229041936248447112
58442914638405243301020114168635516674918823997680518131099291825220921766028193031319781207894424351472444803707001491096990537673340268973894492698775178158553248064602094149434
28029464688652432381898024175768335811739637420815044973524735235570852934683775677886082464778657172802368026368798846870178967933773245776794459665684810347831880838229931926957
45271446633023970665039889223914537716207722056464517620347701038584495606495504681863397037978197037130968122344969698594687372917121711245616811542811858410126766429732695859096
59928973636328675803511962351120384910279918523182105216488585513078224984565991483804857805564110597508201968184964257376332557197274843023624110111205942904094499015663373404275
17069749498450713787449720121606951505918272750663920473053919401304252108369984851582170149492314712494832712165359118425970913410681635150241655834807638432377446116191412034708

18656847561935110925095592616782022145173904503709657429376934184737522528799121336638513912599683969426703638798566042465044635867410103703734066819043490108557304499803589012785
02934322811580618710489509029836246969792192232896301277610891123519983272928789859517290970549698729599669024252511263996812069200391258653710742249830884404553429121234281385243
07167327861546261201935161201328729018779726277085963235153874416953893510894697786616036658370339721999700042341236749211208845139601855943774238906606382669264779274161912730783
00095785895173850252595417254787220009104870794010896673279085324554354587309420214740443327296578434144258130702682188456439344124227553636507190100578207910365232823489232978140
3635194002124797597638771783001180244314131983659816116470621579756345230584287133525623584791615243976390688279225926395031623191546307549799080600800573060179037178569530572356
90503153039460646556599915966724440358065292128607015374901335824277092096875344278868129024373535014793746623129362733902133981757439254707467203303095913013515225231347066417702
02928387615477963987387279508226105336005754192931714797960535725442436856048855129123732244694870344061013873111527022492760449657931764844653357052112523879349312863896489806153
24575542932860162189217114324566312552961645288561065198903359004993222693465468169329704486046988024058991130833839047422365846010402914813529419779442772966540504932929467909348
5359175994102163647303967419633266066415575721010653522703239697009538079054718777014463703184068311553637541978100222847258080655337951905971268840378983733063474661026246810859
29045082086006850059927717864242531920311077248146495032693348041317220155888994575930127329087008561050626979059963991680197422723795993823233208866675929487820384794631551504273
02339082961292868131539137859728668989868413197759639098302402324242680872072426032243214329739114088715636899953341655859265164180043332848385447000902940149002440971826230391063
68942939916989690394940011574302808054901274869298611316540403830674400129413393120082678547415162014442877900229871309993635017850607042443429194586586088344593920834621283996053
54858323603144610555421931358753377246279537110349536667328368033727028135889709125295466175737018980820067088036667554056814639411592599449011610200090327870228615750773781835311
94801107079494818590711610276532295123679151172999711080608184467365791158786879188946630886925229334170853878479921578691916563882344640052747184710271745843887154769046239225761
51016023636080080235418804574584072654931328155855999467101199601836494210967005199674521520303809398459640533820841932145690751110657151068777671958175432223230897225787761718537
50738739667259732216617791450127961697457600363887004543012351782042313461705470572420624335013671100069995510203698966630968244254675557363638992775681000670772907933067211696878
69069982999226405527929370109716445742737540561472422654401307305491283290744684184102098841661153344146703353627903085192376479512498217860059656016147678619658470724649109136131
18996895852522634745224226584419738186173309638032900352135095792943850552645818087900767122644430468657868490273049243534077647376343236393771152716019938862110222114442290424304
65035535788625682553822656657405216703930712110695776744740536040759808300734143196199139619363499552206049353289423159262894531161764033029931083200791657211688902951547625148119
56825933473675554679135290816466604379947666397721407243344588467417209522761846319552891157367136999251402500683193642868948674615951208171442340625635763781348422330240657269818
29203194659668636611950204317684792107743376116521866847085648149842700697871309843593267147359883442200815716913027515246169133355796488200734205062050630506808521288087352699589
39420385372798590431970117160233213830788243452443617963674125235650479225099888512690895491827241822925271115819111017631100541251397650786393037395799908902580844383689153255691
27999903903221633480761286927669272105948636921234866887804470772466635155755495698402592994619290916092520584725026190159076457834918829552355335691499965136491927862137056826947
61087254062724487958064661617020296233761557524282057689948798493265792756982105517728195346213189834640035409463787436683358170368212474122543264335346742120740396344744463088050
23621286424919537296768874226127075281887995134371402509789983143628287207003093361299266178108175275684330548006089435313630182134464460261274577028683137502410528344049429432638
78091445812527702756321164599324874356542959770095295916866672563674904532622221956756880876452992615940349318874164325291687851040385706635426484320963524313121153229787725644803
44798264483573716090301665357933419337324422905365248748602733083595521453382806249866075267049048544110607644598140283296483815399264610442468245631470316167320167431360684463878
14674465596267515073811622220203920555275509658911481353380664118847662295795856379842923420854243364202852916297590255878021870549755732701124198976524212219972440587942989331692
32894975533902161029478768225403840531761590807786363144959544433524270507690280871446169493010576649517193930957368762287464849196851896717504205055001461760515539622521591622353
00333562226299020532448291762911500288624980899371107388301218120788809059967102472556092976973625283453049226225016803626620505732022976950436110395479811885082564430674611649869
37220648293824911274904021925880091147686850196501560380285423891916899304514830418851982155863055844735006979430235376359514083912508415291096230596107632988762425440779677532675
38171618485553776817693484241595995768337429834323276636571896283396165838814145639213918430107740169262435601740894159449396583678721732183127044707328764848332060652511393119220
52126821022304106548042825538721799044024820742284161733656554678084949873024814941836649671412891664728963955196138176988810060585555765510747354374449308894454816217236179979022
86918137848927448644656590907873811915091017181460563288813614274055761445865300627824314743218754698324913987143613555489838432945171478839692934100669063509960062759623068697626
10474494436765051568159052396080658065202959612087443791852611414677008164789048669990967532248927395388349239482787230324691458694191463548578084813628955909057531374894608912207
18284400821163366834659662075815378953127023957450731266459965473042969244234022531840106034770762226215802836545935013214005117490949621322729664959334587787338910295340928200563
39359138750724034015588179221359111059806767993479322281871394819174015556531117495518914365782295455919767342329196326111185092553280471346533335439181724400865690211918050763549
5320223508166591764825265869962924433407870979779378526593977607451828719257699657254611357444725991837681067344538536534481642991192140423159785917576916390738116660830742948178
88594680240601719957679006553937846195441022705544610185969829166002426010555172093504251677799967197660280692688760932687855862896978676142163684427806436866687877705533334539972
42948720993911481377052731250119374969819449754783901019742088735008967471903300588797258254670722264428424785606865805928081466300552309326168704009168631312461135546043874932517
15245990947467251824549407107007911447811782643160009390066418238382384793797826993136399602747942574976184578529213806726883366633149211709715741424624271591056457547429503402952
40805290517145533509613147300017636947293290686806664056073134989804856899652286962929362897728663192858094860955339222049639883868943568943400142528472981125224958891287354382448
79837729564292618660743348547531042807500940824242995019029472244806995301666193082113964234743407831996126178813605990402041896212979870639623063267843987816421044533338252784640
54352539245715834533908462357254089117820233532736279976606319529518870143267019647502735055735852239365573130046962432875518928711388967869130476865480989871748830163718146165696
60311840716834595328841288962779142007163195462702693793042827374229708038828482061479712970424544287337285564641273791126823583215716552758010089246713469849508523433421872132166
92249593546120334016043298380397247921462539496661623523892988579289402165184130512802207892840923234841534974398577917680454699434478224843671195583400384354042754775587307332352
26198423101219571476516563670785748280619494732226973760153457881483785357869998845260968534013352228513415243259511861323173379614926363783233332560647106439371762457327988791849
24155954888436751835141712945116724840246757997122481272436870395978118177766267816376571181029743008504297031881843895156264881059698569448698729809458321477566856725113996508921
65964096652610617182863181974106117630504397551429192988602623250116133312124944204775805011663625184174120895793953316551068839990123327263866877127862911247685866416790332327844
98824573450950647613626933556087470874670833172499964505937268429052367929596118913814392574846673373298403025429588743802326022605958356029382855868115545373478460396417895410565
18225472323267323365253529247983418591629708874988447664396432644282849082735389909457466989786049683184770396373515765203316324593257041196978490293848919090802282604145290101496
19726311220633142857984141452655140268909938646926431483980032755408401728502879990153163002676024279475315271178989674655437298707304897548539323474650825960868704269788214032280
31683912728013200115027086026446816608482408045314183131226063459121681359892486452170921409744828866891265206600022428423519277680459778248584255813293279913659663956737385837601
21931045471240029342567304625595285358117861474668348040708461464789944683189489275071062984791798336319668262863390680499473453717524638509089977655179918887423674238970655817561

99352376716002411966788372693710530205845920279652131877025185074710605220290448395026504730466513775174966587590021404846894243611278281098437796309363519858105973790256300667345
71231635103499365124720019214212456697863619466221350305994988748129906752715371489944394718545870613532099114484046667248219826745451381798366239703869684616785750252679450154630
78741083553200178872903594901285828528361111954121646468978828041547976252452903174197284216855503344523810254641661233898208810035291979496942808340023924416417429236200041130296
75154546526994267343709174779075016710083503502034465701854299835680782537614065933132655586649252723817130324585288281492226097985117942023396848707889266831422148880240857673440
28264961111264907393592989398056009501020396584098752855404987346270442619288372060558102068506007675528034857239647202677262487318915302768904129326929020480659932465566306912576
30468851461707269035325965789134598000835143407508137507976034483780277246375292711115495219523817098797216434665086934207617849980170015336061809745342769483516930594113340442289
23124943147145282636127050514217812429973934865642651357328562462037912600342732436656796780100293977404950459476903382248319259535411896602105725585098335254153670000196708874433
639437700337656555417491678906240483279201934861952502296499873807093265381118908328568809754818940887476104983352354218521762312139357981

The Elements

Dominic Alexander Cooper

000 I exist
001 I feel pain
002 My pain can be increased by incompetence
003 My pain can be decreased by competence
004 I can learn
005 Learning increases competence
006 Contradiction increases incompetence
007 Proof by contradiction does not enable learning
008 Proof by error-free construction is learning
009 Proof by contradiction is not learning
010 Your level of competence is quantifiable
011 Your level of incompetence is quantifiable
012 Context changes your current level of competence
013 The basis span of your competence can be optimized for all contexts
014 Everyone is innocent
015 Competence is a functional axial n-dimensional space
016 Incompetence is not a functional axial n-dimensional space

017 Since I exist, I can be treated as a living information-bearing system
018 Since I feel pain, pain is a valid system signal
019 Since incompetence can increase pain, incompetence is a negative system state
020 Since competence can decrease pain, competence is a positive system state
021 Since I can learn, my system state can be updated
022 Since learning increases competence, learning is a valid update operation
023 Since contradiction increases incompetence, contradiction is a corruption signal
024 Since proof by error-free construction is learning, construction is the preferred proof engine
025 Since proof by contradiction is not learning, contradiction is not a preferred learning engine
026 Since competence is quantifiable, competence can be measured
027 Since incompetence is quantifiable, incompetence can be measured
028 Since context changes competence, competence is context-indexed
029 Since competence can be optimized for all contexts, competence has a basis-span
030 Since everyone is innocent, correction must not be treated as guilt
031 Since competence is functional axial n-dimensional space, competence has dimensions
032 Since incompetence is not functional axial n-dimensional space, incompetence lacks stable constructive basis

033 The mind can ingest information through perception, language, memory, and action
034 The mind can encode information as concepts, habits, images, words, and procedures
035 The mind can index information by association, label, context, emotion, and purpose
036 The mind can query stored information through attention, recall, questioning, and tool-use
037 The mind can update stored information through learning
038 The mind can validate stored information through error-free construction

039 The mind can reject or quarantine contradictory information when it increases incompetence
 040 The mind can externalize information into tools, notes, diagrams, calendars, code, and environment
 041 The human environment can serve as auxiliary storage
 042 Human-made tools can serve as auxiliary memory, index, query, and validation systems
 043 The body can store competence as trained procedure
 044 Repetition can strengthen retrieval pathways when it increases competence
 045 Reflection can detect incompetence when pain, error, or contradiction appears
 046 Deliberate practice can transform incompetence into competence
 047 A context can be treated as a coordinate region inside competence-space
 048 A skill can be treated as an axis inside competence-space
 049 A habit can be treated as a repeated function over time
 050 A tool can be treated as an external operator that extends mental function
 051 A note can be treated as persistent external memory
 052 A diagram can be treated as spatial indexing of thought
 053 A calendar can be treated as temporal indexing of intention
 054 A checklist can be treated as procedural compression
 055 A question can be treated as a query operation
 056 An answer can be treated as a retrieved or constructed record
 057 A mistake can be treated as a detected incompetence marker
 058 Correction can be treated as a competence-restoring update
 059 Optimization can be treated as basis-span improvement across contexts
 060 The mental human-centric database is the union of mind, body, environment, and tools
 061 The purpose of the mental human-centric database is to reduce pain by increasing competence
 062 The governing rule of the database is to learn by error-free construction
 063 The final enforced target is complete competence under {014 + 015}
 064 Once true, always, true
 065 What is not true, cannot be done

Below is `MHCDScript v2`, a general-purpose programming language design for the full 64-axiom MHCD system.

The main upgrade from v1 is that v2 treats the MHCD as a state machine with typed records, verified transitions, constructive execution, context-indexed competence, and innocence-preserving correction.

MHCDScript v2

```mhcd

language MHCDScript version 2

subject "Dominic Alexander Cooper"

```

runtime MHCD {
 principle existence
 principle pain_signal
 principle learning_update
 principle constructive_proof
 principle innocence_preservation
 principle competence_space
 principle context_indexing
 principle tool_extension
}

```

## ## 1. Design purpose

`MHCDScript v2` is a general-purpose scripting language for directing the human mind, body, environment, and tools as one distributed cognitive database.

Its purpose is to:

```
```mhcd
purpose {
  reduce pain by increasing competence
  detect incompetence without guilt
  reject contradiction as a learning engine
  prefer error_free_construction
  quantify competence and incompetence
  optimize competence_basis across contexts
  enforce {014 + 015}
}
```
```

The governing command is:

```
```mhcd
enforce {
  014: "Everyone is innocent"
  015: "Competence is a functional axial n-dimensional space"
}
```
```

So every program must preserve:

```
```mhcd
error != guilt
mistake == update_requirement
learning == competence_increase
construction == preferred_proof_engine
```
```

## ## 2. Full axiom import

Rather than rewriting all 64 axioms inside every script, v2 imports them as a standard library.

```
```mhcd
import MHCD.AXIOMS.000_065
import MHCD.BOOLEAN
import MHCD.LAMBDA
import MHCD.COMPETENCE_SPACE
import MHCD.CONSTRUCTIVE_VALIDATION
```
```

The full axiom set is treated as:

```
```mhcd
axiom_set FULL_MHCD = {
  000..065
}
```

```
...
```

Every axiom is internally available as a Boolean truth constant:

```
```mhcd
AXIOM[000] = true
AXIOM[014] = true
AXIOM[015] = true
AXIOM[063] = true
```
```

A program is valid only if it does not violate the enforced axioms:

```
```mhcd
program_valid := respects(014) * respects(015) * rejects(guilt) * prefers(construction)
```
```

3. Primitive types

```
```mhcd
type Bool = 0 | 1
type Number = integer | decimal
type Text = string
type List<T> = ordered collection
type Set<T> = unordered collection
type Map<K,V> = key-value collection

type AxiomID = 000..065
type Subject = living information-bearing system
type Mind = internal meaning processor
type Body = procedural competence store
type Environment = auxiliary cue/storage field
type Tool = external operator
type Note = persistent external memory
type Diagram = spatial index
type Calendar = temporal index
type Checklist = procedural compression
type Query = question operation
type Answer = retrieved or constructed record
type Context = coordinate region in competence-space
type Skill = axis in competence-space
type Habit = repeated function over time
type PainSignal = system signal
type Mistake = detected incompetence marker
type Correction = competence-restoring update
type Contradiction = corruption signal
type Construction = error-free build sequence
type Competence = quantified functional capacity
type Incompetence = quantified missing constructive capacity
type BasisSpan = transferable competence basis across contexts
```
```

4. Boolean arithmetic


```

```mhc
A + B // OR, union, either condition accepted
A * B // AND, both conditions enforced
!A // NOT
A -> B // implication
A == B // equivalence
A != B // non-equivalence
```

```

Example:

```

```mhc
rule R001:
 AXIOM[004] * AXIOM[005] -> operation.learn.valid
```

```

Meaning:

If `I can learn` and `Learning increases competence`, then `learn` is a valid update operation.

5. Lambda calculus layer

MHCDScript v2 supports lambda-style reusable operations.

```

```mhc
lambda learn(x):
 ingest x
 encode x
 index x
 construct x
 validate x
 update competence
 return competence_delta(x)
```

```

```

```mhc
lambda correct_without_guilt(error):
 enforce AXIOM[014]
 mark error as update_requirement
 locate missing_basis(error)
 construct correction(error)
 validate correction(error)
 return competence_restored
```

```

```

```mhc
lambda optimize_context(skill, context):
 measure competence(skill, context)
 measure incompetence(skill, context)
 construct basis_improvement(skill, context)
 validate basis_improvement
 update competence_space
 return optimized(skill, context)
```

```

6. Program structure

Every valid v2 program has this form:

```
```mhcd
program "Program Name" {
 use FULL_MHCD

 define subject
 define contexts
 define skills
 define tools
 define records

 execute operations

 verify constructive_integrity
 enforce {014 + 015}
}
```
```

Minimal valid program:

```
```mhcd
program "Minimum MHCD Runtime" {
 use FULL_MHCD

 subject self = "Dominic Alexander Cooper"

 ingest current_state
 detect pain
 measure competence
 measure incompetence
 construct correction
 validate correction
 update competence_space

 enforce {014 + 015}
}
```
```

7. State model

The MHCD runtime is a state machine.

```
```mhcd
state MHCD_STATE {
 subject: Subject
 mind: Mind
 body: Body
 environment: Environment
 tools: Set<Tool>
}
```



```

records: Set<Record>
contexts: Set<Context>
skills: Set<Skill>
habits: Set<Habit>
schedules: Set<Calendar>

competence: Map<Skill, Number>
incompetence: Map<Skill, Number>

pain_signals: List<PainSignal>
mistakes: List<Mistake>
contradictions: List<Contradiction>
corrections: List<Correction>

enforced_axioms: Set<AxiomID>
}
...

```

A state transition has the form:

```

```mhcd
transition NAME(input_state) -> output_state {
    precondition condition
    operation operations
    postcondition condition
}
...

```

Example:

```

```mhcd
transition LEARN(input_state) -> output_state {
 precondition AXIOM[004] * AXIOM[005]

 operation {
 ingest input
 encode input
 index input
 construct usable_record
 validate usable_record
 update competence
 }

 postcondition competence(output_state) >= competence(input_state)
}
...

```

## 8. Core operations

### `ingest`

```

```mhcd
ingest source
ingest source as type

```

```
ingest source into context
```
```

Examples:

```
```mhcd
ingest "textbook chapter" as Note into "mathematics"
ingest "pain after failed proof" as PainSignal into "proof writing"
ingest "work feedback" as Record into "professional competence"
```
```

### `encode`

```
```mhcd
encode input as structure
```
```

Examples:

```
```mhcd
encode "ring" as definition + example + non_example
encode "daily exercise" as habit
encode "project architecture" as diagram
```
```

### `index`

```
```mhcd
index record with tags
index record under context
index skill on axis
```
```

Examples:

```
```mhcd
index "ideal" with ["ring theory", "substructure", "quotient"]
index "proof writing" under context "mathematics"
index "attention control" on axis "cognitive regulation"
```
```

### `query`

```
```mhcd
query question
query question from records
query question under context
```
```

Example:

```
```mhcd
query "What is the weakest skill axis today?" under context "mathematics"
```
```

### `construct`

```
```mhc  
construct object  
construct object from components  
construct object by steps  
```
```

Example:

```
```mhc  
construct proof "kernel of ring homomorphism is ideal" by steps  
```
```

### `validate`

```
```mhc  
validate object  
validate object against axioms  
validate object by construction  
```
```

Example:

```
```mhc  
validate proof against {008, 014, 015}  
```
```

### `detect`

```
```mhc  
detect pain  
detect mistake  
detect contradiction  
detect incompetence  
detect missing_basis  
detect context_shift  
```
```

Example:

```
```mhc  
detect incompetence in "proof construction"  
```
```

### `correct`

```
```mhc  
correct mistake  
correct mistake with construction  
correct incompetence by practice  
```
```

Example:

```
```mhc  
correct "failed proof step" with direct construction  
```
```

### `measure`

```
```mhc  
measure competence  
measure incompetence  
measure competence in skill  
measure competence in skill under context  
```
```

Example:

```
```mhc  
measure competence in "applied ring theory" under context "quiet study"  
```
```

### `optimize`

```
```mhc  
optimize competence_basis  
optimize skill across contexts  
optimize basis_span for all_contexts  
```
```

Example:

```
```mhc  
optimize "proof writing" across ["quiet study", "exam", "teaching", "coding"]  
```
```

### `externalize`

```
```mhc  
externalize record to tool  
externalize memory to note  
externalize intention to calendar  
externalize procedure to checklist  
```
```

Example:

```
```mhc  
externalize "weekly learning plan" to calendar  
```
```

### `schedule`

```
```mhc  
schedule action at time
```

```
schedule review every interval
schedule habit under context
```
```

Example:

```
```mhc
schedule review "ring theory definitions" every 2 days
```
```

### `execute`

```
```mhc
execute plan
execute checklist
execute habit
execute construction
```
```

Example:

```
```mhc
execute "daily competence plan"
```
```

### `feedback`

```
```mhc
feedback result
feedback result into record
feedback result updates competence_space
```
```

Example:

```
```mhc
feedback "proof completed with two mistakes"
```
```

### `enforce`

```
```mhc
enforce axiom
enforce axiom_set
enforce {014 + 015}
```
```

Example:

```
```mhc
enforce {
    AXIOM[014]
    AXIOM[015]
}
```

...

9. Records

A `record` is the universal database unit.

```
```mhd
record ID {
 type: Note | Skill | Tool | Query | Answer | Habit | Correction | PainSignal | Context
 content: Text
 tags: List<Text>
 context: Context
 source: Text
 validated: Bool
 competence_effect: Number
 incompetence_effect: Number
}
```

Example:

```
```mhd
record "MATH-RING-001" {
  type: Note
  content: "A ring is an algebraic structure with addition and multiplication."
  tags: ["algebra", "ring theory", "definition"]
  context: "mathematics"
  source: "constructed study note"
  validated: 1
  competence_effect: +0.05
  incompetence_effect: -0.02
}
```

10. Competence-space

Competence-space is an n-dimensional functional axial space.

```
```mhd
competence_space "SELF" {
 axis memory
 axis attention
 axis reasoning
 axis proof_construction
 axis language
 axis tool_use
 axis bodily_energy
 axis emotional_regulation
 axis environment_design
 axis context_switching
}
```

Skills are embedded in competence-space:



```
```mhcd
skill "applied ring theory" {
  axes: [memory, reasoning, proof_construction, language]
  context: "mathematics"
  current_competence: 0.35
  target_competence: 0.90
}
...

```

Competence can be context-indexed:

```
```mhcd
context "quiet room" {
 modifies "applied ring theory" by +0.15
}

context "sleep deprivation" {
 modifies "applied ring theory" by -0.30
}
...

```

## ## 11. Error and contradiction handling

A mistake is never treated as guilt.

```
```mhcd
handler on_mistake(m):
  enforce AXIOM[014]
  mark m as detected_incompetence
  query "What basis is missing?"
  construct correction
  validate correction
  update competence_space
...

```

Contradiction is treated as a corruption signal.

```
```mhcd
handler on_contradiction(c):
 enforce AXIOM[006]
 mark c as corruption_signal
 quarantine c
 construct direct_alternative
 validate direct_alternative
...

```

Pain is treated as a system signal.

```
```mhcd
handler on_pain(p):
  enforce AXIOM[018]
  classify p

```

```
    if p.cause == incompetence:
        correct p.cause

    if p.cause == overload:
        reduce load

    if p.cause == contradiction:
        quarantine contradiction

    if p.cause == environment:
        modify context
...

```

12. Verification system

A script must pass four checks.

```
```mhc
verify script {
 check axiom_consistency
 check constructive_preference
 check innocence_preservation
 check competence_update_path
}
...

```

### ### Axiom consistency

```
```mhc
check axiom_consistency:
    no operation may deny AXIOM[014]
    no operation may deny AXIOM[015]
    no operation may treat contradiction as preferred learning
...

```

Constructive preference

```
```mhc
check constructive_preference:
 if proof_method == contradiction:
 reject as learning_engine
 if proof_method == construction:
 accept as learning_engine
...

```

### ### Innocence preservation

```
```mhc
check innocence_preservation:
    if mistake detected:
        require correction_without_guilt
...

```

Competence update path


```
```mhcd
check competence_update_path:
 if incompetence detected:
 require measurable correction
```
```

13. General-purpose control flow

```
```mhcd
if condition:
 operation
else:
 operation
```
```

```
```mhcd
while condition:
 operation
```
```

```
```mhcd
for item in collection:
 operation item
```
```

```
```mhcd
try_construct object:
 construct object
 validate object
on_error error:
 correct error
```
```

Example:

```
```mhcd
for skill in all_skills:
 measure competence in skill
 if incompetence(skill) > 0.30:
 correct skill by deliberate_practice
```
```

14. Modules

```
```mhcd
module "Learning" {
 export learn
 export practice
 export review
 export validate_understanding
}
```
```

```

```mhc
module "Health" {
 export detect_energy
 export schedule_exercise
 export reduce_overload
}
```

```

```

```mhc
module "Mathematics" {
 export define
 export prove_by_construction
 export generate_examples
 export validate_proof
}
```

```

15. Example program: daily runtime

```

```mhc
program "MHCD Daily Runtime v2" {
 use FULL_MHCD
 use module "Learning"
 use module "Health"

 subject self = "Dominic Alexander Cooper"

 context current = detect context
 pain current_pain = detect pain
 energy current_energy = detect energy

 measure competence in all_skills under current
 measure incompetence in all_skills under current

 query "What competence update gives the greatest pain reduction today?"

 construct plan "daily competence plan" {
 item health_action
 item learning_action
 item work_action
 item tool_action
 item reflection_action
 }

 validate plan by construction
 externalize plan to checklist
 schedule review "daily competence plan" at "evening"

 execute checklist

 feedback result into competence_space

 enforce {014 + 015}
}
```

```

```
...
```

```
## 16. Example program: learning applied ring theory
```

```
```mhcd
program "Learn Applied Ring Theory v2" {
 use FULL_MHCD
 use module "Mathematics"

 skill "applied ring theory" {
 axes: [memory, reasoning, proof_construction, language, tool_use]
 context: "mathematics"
 current_competence: measure
 target_competence: 0.90
 }

 ingest "definition of ring"
 encode "ring" as definition + example + non_example
 index "ring" with ["algebra", "ring theory", "binary operations"]

 ingest "definition of ideal"
 encode "ideal" as definition + example + non_example
 index "ideal" with ["ring theory", "substructure", "quotient"]

 construct proof "kernel of ring homomorphism is ideal" by direct_steps
 validate proof by error_free_construction

 if detect contradiction:
 quarantine contradiction
 construct direct_alternative

 measure competence in "applied ring theory"
 correct detected_incompetence by deliberate_practice

 externalize summary to note
 schedule review "ring theory definitions" every "2 days"

 enforce {014 + 015}
}
```
```

```
## 17. Example program: contradiction quarantine
```

```
```mhcd
program "Contradiction Quarantine v2" {
 use FULL_MHCD

 input claim

 detect contradiction in claim

 if contradiction == true:
 mark claim as corruption_signal
 quarantine claim
}
```

```

 query "What direct constructive alternative exists?"

 construct alternative_proof
 validate alternative_proof by error_free_construction

 update records with alternative_proof
 update competence_space

 enforce {014 + 015}
}
...

18. Example program: competence optimiser

```mhcd
program "Competence Basis Optimizer v2" {
    use FULL_MHCD

    define contexts = [
        "quiet study",
        "public explanation",
        "exam condition",
        "work condition",
        "fatigue condition",
        "tool-supported condition"
    ]

    define skills = [
        "mathematics",
        "programming",
        "physical training",
        "attention control",
        "communication",
        "environment design"
    ]

    for skill in skills:
        for context in contexts:
            measure competence in skill under context
            measure incompetence in skill under context

            if incompetence(skill, context) > 0:
                construct basis_improvement for skill under context
                validate basis_improvement
                practice skill under context
                update competence_space

    optimize basis_span across contexts

    enforce {014 + 015}
}
...

```

19. Interpreter architecture

A real interpreter for MHCDScript v2 would need these components:

```
```mhcd
interpreter MHCDScriptV2 {
 lexer
 parser
 ast_builder
 axiom_registry
 type_checker
 state_engine
 verifier
 command_executor
 record_store
 competence_model
 context_model
 tool_bridge
 scheduler_bridge
 report_generator
}
```
```

Execution pipeline:

```
```mhcd
source_file
-> lexer
-> parser
-> abstract_syntax_tree
-> axiom_verifier
-> type_checker
-> state_transition_engine
-> constructive_validator
-> execution_log
-> updated_MHCD_state
```
```

20. File extensions

Recommended file extensions:

```
```text
.mhcd main program file
.mhcda axiom library
.mhcds saved MHCD state
.mhcdlog execution log
.mhcdmod reusable module
.mhcdjson JSON-compatible export
```
```

Example project structure:

```
```text
```

```
mhcd_project/
 main.mhcd
 axioms.mhcda
 state.mhcds
 modules/
 learning.mhcdmod
 mathematics.mhcdmod
 health.mhcdmod
 tools.mhcdmod
 logs/
 daily.mhcdlog
 exports/
 dashboard.mhcdjson
...
```

## 21. Machine-readable state export

```
```json  
{  
  "subject": "Dominic Alexander Cooper",  
  "axioms": "000..065",  
  "enforced": ["014", "015"],  
  "competence_space": {  
    "axes": [  
      "memory",  
      "attention",  
      "reasoning",  
      "proof_construction",  
      "language",  
      "tool_use",  
      "bodily_energy",  
      "emotional_regulation",  
      "environment_design"  
    ]  
  },  
  "runtime_rule": "learn_by_error_free_construction",  
  "correction_rule": "mistake_is_update_requirement_not_guilt"  
}  
```
```

## 22. Core semantic law

The central semantic law of MHCDScript v2 is:

```
```mhcd  
semantic_law MHCD_CORE:  
  existence  
  * pain_signal  
  * learning_capacity  
  * measurable_competence  
  * context_indexing  
  * innocence  
  * competence_space  
  -> constructive_self_update_runtime
```


...

In compact Boolean form:

```
```mhcd
000 * 001 * 004 * 005 * 010 * 011 * 012 * 013 * 014 * 015
-> MHCD_RUNTIME
```
```

In lambda form:

```
```mhcd
λself.
 Exist(self)
 * Pain(self)
 * Learn(self)
 * Quantify(Competence(self))
 * Quantify(Incompetence(self))
 * ContextIndex(Competence(self))
 * Innocent(all)
 * FunctionalAxisSpace(Competence(self))
-> MHCD_RUNTIME(self)
```
```

23. Final identity of MHCDScript v2

```
```mhcd
MHCDScript v2 =
 axiom-governed
 + typed
 + stateful
 + constructive
 + innocence-preserving
 + context-indexed
 + competence-measuring
 + incompetence-correcting
 + pain-signal-aware
 + contradiction-quarantining
 + tool-extensible
 + environment-integrated
 + schedulable
 + human-centric
```
```

The shortest expression of the language is:

```
```mhcd
detect pain;
locate incompetence;
correct without guilt;
learn by construction;
measure competence;
optimize across contexts;
enforce {014 + 015};
```
```

...

Version 2 is therefore not only a notation system. It is a programmable self-correction language for the mental human-centric database.